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Chronic intracranial recordings in the basal ganglia reveal patient-specific circadian rhythms in Parkinson's disease

Coralie de Hemptinne

coralie.dehemptinne@neurology.ufl.edu

Jackson Cagle
University of Florida https://orcid.org/0000-0002-1955-8050

Tiberio de Araujo
University of Florida

Kara Johnson
University of Florida

Jun Yu
University of Florida https://orcid.org/0000-0002-6222-0717

Lauren Fanty
University of Florida

Filipe Sarmento
University of Florida https://orcid.org/0000-0001-6972-6414

Simon Little
University California San Francisco https://orcid.org/0000-0001-6249-6230

Michael Okun
University of Florida Fixel Institute for Neurological Diseases https://orcid.org/0000-0002-6247-9358

Joshua Wong
University of Florida Department of Neurology

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Abstract

Adaptive deep brain stimulation (aDBS) is a promising therapeutic approach for subjects with Parkinson's disease (PD) using neural signals as feedback control to adjust the level of therapy. However, the circadian fluctuation of neural signals may alter these control signals and possibly result in suboptimal therapy. We report a large retrospective cohort including 119 subjects with PD (165 hemispheres) with globus pallidus internus (GPi) and subthalamic nucleus (STN) activity recorded chronically in the home environment. Both GPi and STN activity was significantly altered by circadian rhythms. The direction of the circadian rhythm (increase versus decrease at night) depended on the frequency band in both the GPi and STN. The use of long-acting extended released dopaminergic medications significantly impacted the circadian rhythms in the GPi target. These findings demonstrate the variability of chronic circadian rhythms and suggest that aDBS will likely need to account for these patient-specific fluctuations to ensure successful implementation in real-world environments.

Introduction

Parkinson's disease (PD) is a neurodegenerative movement disorder characterized by the loss of dopaminergic neurons that leads to the development of motor and nonmotor symptoms. Deep brain stimulation (DBS) is an established therapy to treat motor symptoms in selected subjects with medication-refractory PD[1,2]. This invasive neurosurgical procedure consists in delivering electrical stimulation to the deep brain nuclei through electrodes implanted in the basal ganglia nuclei. In PD, DBS is commonly targeted to either the globus pallidus internus (GPi) or the subthalamic nucleus (STN), two key nodes in the basal ganglia-thalamo-cortical networks underlying PD pathophysiology[3].

Currently, DBS therapy for PD delivers stimulation continuously to disrupt pathological neural activity. However, recent research has increasingly focused on the development of adaptive DBS (aDBS) algorithms, which use neural signals as feedback control to increase or to decrease stimulation based on the presence of pathological activity[4]. The development of aDBS has been recently facilitated by advancements in DBS devices to support bidirectional sensing-stimulation devices, enabling simultaneous stimulation and recording of local field potential (LFP), both in the clinic and the home environment. LFP activity in the beta frequency band (13–30 Hz) has been most commonly used as the aDBS feedback control signal[5,6]. USe of this signal has been largely based on established evidence that beta activity in both the GPi and STN correlate with PD rigidity and bradykinesia [7–10]. Experimental use of aDBS has been shown to improve symptoms with less stimulation-induced side effects and has been shown to improve battery life[5,6,11,12]. Narrow band gamma activity (70-90Hz) has been suggested as a marker of the 'on medication PD state' and the hyperkinetic state of dyskinesia[13], and has aslo been successfully used as feedback signals for aDBS. Chronic use of such algorithms over weeks, in patients' home environment, has been shown to potentially improve on-time state and quality of life in most subjects studied[14]. Therefore, aDBS is an approach to improve the efficacy and efficiency of DBS for PD.

Although encouraging, implementation of aDBS has been found to be more challenging than expected. One factor that has added complexity to stimulation parameterization is the presence of natural fluctuation of biomarkers due to sleep/wake cycles. Indeed, chronic brain recordings have shed light on rhythmic activity, more specifically circadian rhythms, that could potentially alter the target of aDBS algorithms[15]. Circadian rhythms are endogenous fluctuations that orchestrate a wide range of complex physiological and behavioral processes across the ~ 24-hour sleep/wake cycle[16]. Dopamine contributes to sleep/wake cycle regulation[17–19] via pathways from the hypothalamus to the mesolimbic pathways[20–23]. Circadian rhythms disturbances have been observed in PD, resulting in a broad range of symptoms, including as sleep-wake disorders, autonomic dysregulation, temperature imbalance, and motor fluctuations[24].

Recent studies have revealed that the neural activity in the GPi and STN in subjects with PD exhibited circadian patterns[15] and these fluctuated during different stages of sleep[25–33]. A recent study evaluating chronic STN LFP recordings in the home environment has revealed beta power circadian fluctuations, with a consistent reduction of beta power during sleep[25]. This result supported prior findings from acute STN recordings, performed in-clinic using externalized electrodes over 1 night, and showing that, on average, beta (13-30Hz) and gamma (30-60Hz) power in the STN decreased, while delta (0-3Hz), theta (3-7Hz), and alpha (7-13Hz) power increased during non-rapid-eye-movement (NREM) sleep when compared to wakefulness and rapid-eye-movement (REM) sleep. While studies investigating circadian rhythm via subcortical electrophysiological signals have largely focused on STN, two recent studies by Yin et al have focused on GPi recordings in PD. When compared to Huntington's disease and dystonia Yin et al found that, similar to STN, pallidal beta and gamma activity was reduced in NREM compared to REM sleep and awake state, while delta, theta, and alpha power were increased during NREM sleep[29,30]. However, overall, most studies have been conducted in small cohorts, and have focused on STN activity, recorded in the off-medication state, through externalized electrodes, and performed in controlled environments. Therefore, how circadian rhythm could modulate basal ganglia activity, in particular in GPi, and in patient's naturalistic environment remains largely unknown.

The objective of this study was to characterize circadian patterns of neural activity recorded in the GPi and STN in a large cohort of subjects with PD using longitudinal, at-home recordings via a chronically implanted neurostimulator. An increased understanding of neurophysiologic biomarkers of symptoms and their association with circadian patterns is required to develop future aDBS paradigms that may enhance the treatment of PD symptoms.

Results

In this retrospective study, 119 patients (165 unique hemispheres) were recruited from an IRB approved movement disorders database (UF INFORM). Subjects selected were diagnosed with PD and implanted with the Medtronic Percept PC neurostimulators attached to unilateral or bilateral DBS leads in either the GPi (96 subjects, 135 hemispheres) or the STN (23 patients, 30 hemispheres). A total of 88 males and 29

females were included in this study, with an average age at surgery of 67.8 ± 9.0 (mean ± std) years and an average disease duration of 13.8 ± 6.5 (mean ± std) years. Table 1 summarizes the subjects characteristics for the cohort.

In each subject, neural activity was recorded chronically in participants' home environment while under both therapeutic chronic medication and stimulation conditions. Therefore, for each subject and each hemisphere, we selected the first chronic neural recording, following DBS implantation, lasting at least 5-consecutive-day worth of data and out stimulation settings through the recordings. The average recording duration was 17.8 ± 19.7 (mean ± std) consecutive days commencing at 13 ± 35 days after pulse generator device implantation, which happens 4 weeks after DBS lead implantation at our center. The bipolar contact pair used for sensing was determined by the contact(s) used for continuous stimulation therapy. The frequency of oscillatory activity sensed chronically was selected for each subject by the clinician during DBS programming sessions. The clinicians typically selected the most prominent peak on the power spectral density generated after a short survey recording. The neural frequency tracked ranged from 5.86 Hz to 67.38 Hz across this cohort. For analysis, frequencies were grouped in bands defined as theta/alpha (< = 12 Hz), low-beta (> 12 and < = 20 Hz), high-beta (> 20 and < = 30 Hz), and gamma (> = 31 Hz). In GPi subjects, the theta/alpha band and low-beta band were most commonly reordered (46.7% and 37.0% respectively), while in STN the low-beta band was most commonly recorded (63.3%). In some participants, the gamma band was tracked either due to the presence of a physiology peak or documentation of dyskinetic symptoms.

In addition to sensing configuration, stimulation settings were also extracted for the analysis since most patients were receiving continuous high-frequency stimulation therapy during the chronic recordings. Therapy settings varied across subjects, with stimulation pulse width ranging from 60 to 90 μsec, the stimulation frequency ranging from 110 Hz to 180 Hz, and stimulation amplitude ranging from 0 mA to 4.5 mA.

Beta power was usually reduced at night but could also be increased, especially in GPi

Figure 1A shows chronic GPi high beta (24.41 Hz ± 2.50 Hz) activity recorded in an individual subject over 5 consecutive days. Power fluctuations were observed with a consistent reduction during night-time. The reduction manifested around midnight. Vertical lines indicate the time of the events marked, and ranged from 7am to midnight (EDT), corresponding to a period of higher beta power during the daytime period. The polar plots illustrate the circadian rhythms, showing that beta power was higher during the day (when the events were marked). Interestingly, an increased power at night was also observed in other patients. Figure 1B shows an example of beta (19.53Hz ± 2.50Hz) in a different patient with GPi recordings, consistently increased at night and decreased during the day when the subject marked events. The normalized power collected in each GPi (N = 135) plotted over 24h and sorted by highest power in daytime, is shown on the circadian heatmap (Fig. 1C). Visual inspection shows the presence of a strong circadian rhythm in most recordings. We found a decreased power at night in 57.8%

of the recordings (78 hemispheres) while an increased power was observed in 28.9% (39 hemispheres). In 18 hemispheres (13.3%), there was no statistically significant difference between day and night (see Methods).

As expected, STN power was more consistently reduced at night compared to daytime when compared to the GPi (76.7%, 23 hemispheres), with only 5 hemispheres (16.7%) showing increased power at night-time. Two hemispheres (6.7%) exhibited no statistically significant difference between day and night-time. Figure 2 shows individual examples of STN beta power which was reduced at night (**A**), as well as STN beta power increased at night (**B**). The STN power fluctuation of each recording (n = 30) is shown on the circadian heatmap (Fig. 2C).

Basal ganglia circadian rhythm depends on the frequency band recorded.

Given the variability in circadian rhythm (increase vs decrease power at night), especially in GPi, we investigate the effects of sensing frequency. For each recording, the change in power between day and night was calculated and plotted against the center of the sensing frequency (Fig. 3A**/D**).

In GPi, non-parametric tests within each canonical spectral band (theta/alpha, low-beta, high-beta, and gamma) indicated that changes in neural signals between day and night were significantly different from 0 (no-change) in both high-beta (median = 1.29, $p = 0.044$, FDR corrected) and gamma (median = 1.43, $p = 0.013$, FDR corrected). The positive median values indicate that GPi power in high-beta and gamma bands were more likely to decrease at night. Alpha and Low-Beta power were either decreased or increased during night-time, as indicated by a median non-significantly different from 0 (alpha, $p = 0.576$ and Low-Beta $p = 0.291$, FDR corrected). A one-way ANOVA to compare across spectral bands revealed a significant effect of sensing frequency in GPi ($F = 6.42$, $p < 0.001$). Post-hoc analysis with Tukey's HSD showed that the circadian rhythm in the theta/alpha band showed more increased in night-time than high-beta ($q=-0.81$, $p = 0.023$) and gamma ($q=-1.31$, $p = 0.002$), and low-beta showed more increased in night-time than gamma ($q=-1.07$, $p = 0.019$) (Fig. 3B).

In STN, non-parametric tests showed that low-beta (median = 1.29, $p = 0.033$, FDR corrected) and high-beta (median = 1.29, $p = 0.022$, FDR corrected) were significantly different from 0. The positive median revealed that STN power in low- or high-beta bands was more likely to decrease at night, while theta/alpha power was either increased or decreased. A one-way ANOVA to compare across spectral bands indicated a significant effect of sensing frequency ($F = 4.11$, $p = 0.014$). Post-hoc analysis with Tukey's HSD showed that the circadian rhythm in high-beta decreased more in night-time than theta/alpha ($q=-1.71$, $p = 0.013$) (Fig. 3E).

These results suggest that different frequency bands of brain activity exhibited circadian rhythms of different magnitudes and directions. In both targets, high-beta power was more consistently reduced at night, while theta/alpha power was either increased or reduced at night. However, low-beta band, especially in the GPi, was more likely to increase at night (32.6%) than in STN (11.1%).

Extended-release medications modulate circadian rhythm.

Medications and stimulation are known to modulate basal ganglia activity in PD, especially the beta band. Therefore, a general linear model (GLM) was built to test the effects of stimulation parameters and medications on the direction of beta power (13-30Hz) circadian rhythms in GPi and STN. In particular, we investigated whether the change in beta power (dependent variable) during daytime versus night-time was significantly modulated by the following independent variables: the levodopa equivalent daily dose (LEDD), the use of medications (dopaminergic extended-release, benzodiazepines, and melatoninergic medications), the total energy delivered (TEED) (a measure of the amount of stimulation based on amplitude, pulse width, and frequency), the contact used for stimulation (dorsal/ventral), and the hemisphere recorded (left/right).

Table 2 summarizes the results of the GLM. The model indicated a significant effect of extended-release dopaminergic medications on beta power circadian rhythm ($z = 2.46$, $p = 0.016$), while all other variables (stimulation parameters and other medications) did not significantly affect the change in beta power during daytime versus night-time. In GPi, the group of subjects using extended-release dopaminergic medication showed a statistically significant greater decrease in power at night compared those who were not taking extended-release medication ($W = 112$, $p < 0.001$) (Fig. 3C). Interestingly, the GLM built for STN recordings did not reveal any independent variable that contributed significantly to the direction of circadian rhythm in STN. The use of extended-release dopaminergic medication did not influence the direction of circadian rhythm in STN ($z = 0.61$, $p = 0.540$).

Although the TEED did not significantly affect beta power circadian rhythms at the group level, we observed that stimulation parameters could affect circadian rhythms within individual subjects. In particular, larger stimulation amplitude might reduce the range of neural signal fluctuations as shown in **Figure S1A**. However, the influence of stimulation varies considerably across patients.

Discussion

This retrospective study characterizes the effect of circadian rhythm on PD GPi and STN neurophysiology, using a commercially available device in a large sample size. IThe data revealed that 1) both GPi and STN power is significantly modulated by circadian rhythm in most PD subjects, 2) although power was mainly reduced at night, increased power was also found, especially in GPi (28%), 3) Both the frequency of the power band recorded, and the use of long-acting levodopa medications influenced the neurophysiological circadian rhythm.

The large chronically recorded dataset revealed that basal ganglia activity was strongly modulated by circadian rhythm, and that the frequency of subcortical power recorded could significantly affect the direction of the spectral power changes (increase or decrease) between day and night-time. Previous studies exploring circadian changes in STN LFP[15,25,27,28,34], have shown that different canonical spectral bands behave differently during sleep with a general trend of slowing of brain oscillations, with increased

low frequency oscillatory power (< 12Hz) and decreased beta and gamma power (> 13Hz) at night. Although our STN sub-analysis largely matches these previous STN findings, the GPi findings indicate more variability in pallidal circadian rhythm, especially in the low beta band which was increased at night in ~ 1/3 of the recordings. The increase in power at night, especially in low beta power, could be explained by the large spectral window of the recording. Indeed, the neuronal signal recorded chronically is set at a preselected frequency with a 5Hz window (± 2Hz). Therefore, it is possible that alpha power (~ 10Hz) contaminated recordings at 13Hz. However, our results showed increased power at night even in frequencies above 14Hz, which are unlikely to be contaminated by alpha power (Figs. 1 and 2).

Contrary to most studies on sleep in PD, in our study, the brain activity was collected chronically while the subjects were on typical medication dosages and on stimulation. Thus, to further understand this intriguing result, especially in the beta band, we investigated the effect of both medication and stimulation on circadian beta power. We found that, in GPi, the use of extended-release levodopa modulated the direction of power changes. In particular, a decrease in beta power at night was more likely observed in subjects treated with long-acting levodopa medications when compared to those who were not taking extended-released medication. The use of long-acting levodopa medications could suppress beta power at night-time similar to the daytime[35–37], thus masking the emergence of pathological beta power at night as seen in patients who were not taking extended released medication. Interestingly, the effect of long-acting levodopa was not significant in STN, although this is likely due to lack of power in this subgroup.

Our observations on impact of circadian rhythm on basal ganglia activity has important potential implications for the implementation of aDBS therapy in PD. Currently, aDBS algorithms utilize a single or double threshold approach which is centered around the pathological beta signal in basal ganglia circuitry. The aDBS control system will increase the therapy level when the tracked signal is high and reduce therapy level when the tracked signal is low. Our study has revealed that neural signals in basal ganglia, specifically in the beta-band, had different trends in different subjects. The aDBS therapy may be set to stay at higher duty cycle if the threshold is built on daytime recordings for patients who have significant increase of power at night-time and vice versa. This challenge may result in either unintended additional battery drains or suboptimal therapy in the night-time. In addition, our analysis also indicated that the medication usages may contribute to changes in circadian rhythm, meaning that aDBS algorithm may need re-calibration for night-time therapy when medications, specifically the long-acting levodopa medications, changed. Thus, we caution that while aDBS can improve the responsiveness of neuromodulatory therapy in PD to adapt to activities of daily life, it should not be implemented as a one-size-fits-all approach. We would argue that neuromodulation will need to be tailored to each individual's neural signature for optimal therapeutic benefit and that circadian patterns are an important consideration given that many people spend ~ 1/3 of their time in a sleep state.

It is important to highlight several limitations of our study. As a retrospective study, we did not have wearable sleep data or formal sleep diaries to confirm the sleep/wake states in all subjects and instead we developed a novel automated circadian detection algorithm based on the chronic neural recordings.

However, several previous studies have demonstrated the feasibility of characterizing sleep based on subthalamic and pallidal signals [29,34,38–40]. We validated our algorithm in subjects who had marked event that we used as a surrogate for wakefulness. Additionally, we did not have real-time measurements of the patients' symptoms, disease states, and sleep stages at the time of recordings. However, future studies could incorporate such measurements to better understand the circadian modulation of pallidal signals in greater detail. Similarly, since the recordings were collected from subjects receiving continuous high-frequency DBS and effective DBS therapy has been demonstrated to change LFP signals including during sleep[15], the effect of circadian rhythm could also have been influenced by the impact of the therapeutic stimulation. Although the total electrical energy delivered (TEED) did not significantly explain the variability of GPi beta oscillation circadian rhythm in patients with PD, we observed within-subject effects of changing stimulation parameters on circadian rhythms (Figure S1A). It is thus possible that an important factor is the therapeutic effect of each stimulation parameter, which unfortunately is not available in the current retrospective dataset, as opposed to the simple TEED measurements.

Future clinical studies will need to better characterize the awake/asleep states, sleep stages, PD symptoms, and related parasomnias. These findings could be used to help guide the use of circadian rhythmicity in aDBS strategies.

We conclude that there is a circadian rhythm in neural signals recorded chronically from both GPi and STN in PD, and that the directionality and amplitude of the changes depend on the frequency tracked and the use of long-acting levodopa medications. Application of aDBS will need to account for circadian rhythms and factor them into the algorithms to ensure successful implementation in real world environments.

Methods

Study Participants and Data Collection:

This retrospective study was performed at the Norman Fixel Institute for Neurological Diseases at the University of Florida (UF). All subjects included (117) provided informed consent, in accordance with the Declaration of Helsinki, to participate in the UF INFORM Database, a clinical-research database that hosts subjects' clinical records and neural recordings[41]. The study was approved by UF Institutional Review Board (IRB) to request and process data from UF INFORM Database. The inclusion criteria were individuals diagnosed with PD by a movement disorders-trained neurologist and who received DBS therapy with the sensing-enabled Medtronic Percept PC neurostimulator with unilateral or bilateral electrode implanted in the GPi or STN.

Retrospective data were extracted from the UF INFORM Database, including brain recording (average power every 10min), therapy settings at recording, medications, and preoperative symptom assessments. Chronic sensing was usually enabled if therapy configuration was one of the sensing-

available configurations, and the neural signal recorded was usually selected based on the most prominent spectral peak shown during baseline surveys[42]. As a retrospective study, wearable devices and sleep diaries were not available. However, the Percept DBS device allows subjects to indicate specific 'events' or symptoms (i.e 'medication intake, 'presence of dyskinesia') that might be relevant for their clinical care. Therefore, the time of each event marked during each recording was extracted and used as a proxi for wakefulness. For each patient, the medications dose and schedule, were used to calculate the total Levodopa Equivalent Daily Dose (LEDD) and classified into common medication classes, namely the immediate release dopaminergic medication, the extended-release dopaminergic medication, melatoninergic medication, and Benzodiazepines. A total of 165 recordings were included in this study.

Data Processing

All clinical DBS programming session files retrieved from the UF INFORM Database were imported into the Brain Recording Analysis and Visualization Online (BRAVO) Platform[43]. Therapy configurations and available neural recordings were sorted chronologically. Some subjects have significantly more recordings than others, so only one recording from each unique hemisphere was included in final analysis to ensure all subjects contributed equally. For each patient, the first chronic neural recording with more than 5 days of data for each available hemisphere was extracted. All neural signals were z-score normalized using ± 12 hours of recordings to eliminate non-stationary slow-wave drifts and allow group analysis. All neural recordings were time-aligned on a 24-hour clock based on Eastern Daylight-saving Time (EDT, GMT-04:00). An automated circadian detection algorithm was then implemented to detect day vs night period. First, neural recordings were binned into 2-hour windows with 90-minute overlap to calculate the mean and 99.9% confidence interval of neural signals at each 30-minute mark for each patient. Second, the bins with maximum average neural signal and minimum neural signal were extracted. All bins that overlap with the 99.9% confidence interval of the maximum bin were considered as the "High Neural Signal" period and all bins that overlap with the 99.9% confidence interval of the minimum bin were considered as the "Low Neural Signal" period. The time of 'low' and 'high' neuronal signal periods was used to determine when each period occurred (day vs night). Finally, in a subset of recordings in which events were marked (61%), the time of each events marked by the subjects during those 2 periods was extracted, and used to confirm that patient was awake during the day. The effect of circadian rhythm was calculated as the mean normalized power during day period across all recorded days subtracted by mean normalized power during night period across all recorded days.

Statistical Methods

The circadian rhythm was measured as the normalized difference between the Awake period and Asleep period by subtracting the mean asleep period neural signals from mean awake period neural signals. For each patient, the spectral bands recorded by the neurostimulator were extracted and categorized into "Theta/Alpha" (0-12Hz), "Low Beta" (13-20Hz), "High Beta" (21-30Hz), and "Gamma" (> 31Hz) groups.

One-way ANOVA was used to examine the effect of sensing frequency on circadian rhythm. Post-hoc pairwise comparison between each group were performed using Tukey's test if a significant effect was found from the ANOVA. Non-parametric Mann-Whitney U test, with multiple comparison corrected with false-discovery rate (FDR), was applied within each canonical sensing frequency group to identify if the trend of circadian rhythm is increasing in daytime or decreasing in daytime by testing against 0. In addition, other therapeutic variables such as therapy frequency, therapy pulse width, and therapy amplitude were extracted for the corresponding chronic neural recordings. Medication classes such as dopaminergic and nondopaminergic medication were included along with the therapeutic configurations and sensing frequency in a generalized linear model (GLM) to identify the variables that significantly contributed to the circadian rhythm. The GLM was designed with the change in Beta band (13-30Hz) neural signals as the dependent variable. Independent variables for the GLM included the therapy hemisphere (left/right), therapy contact (dorsal/ventral), TEED[44], LEDD, levodopa extended-release, benzodiazepines, and melatoninergic medications (yes = 1, no = 0).

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Tables

Table 1
Patient demographic, clinical characterization and recording settings.

	GPi	STN	Total	P-value GPi vs STN
Demography				
total participants	96 (135 hemispheres)	23 (30 hemispheres)	119 (165 hemispheres)	
gender	M: 73 (76%) F: 23 (24%)	M: 16 (70%) F: 7 (30%)	M: 89 (75%) F: 30 (25%)	
age at surgery (year)	67.9 ± 9.4	67.8 ± 6.9	67.9 ± 8.9	0.563
disease duration (year)	13.7 ± 5.8	14.2 ± 9.0	13.7 ± 6.4	0.513
Symptoms and Therapy Characteristics				
Preoperative UPDRS	35.4 ± 12.5	37.6 ± 13.5	35.8 ± 12.7	0.492
TEED (μj)	129 ± 111	60 ± 74	118 ± 105	0.037
LEED (mg)	1282 ± 686	1033 ± 486	1234 ± 658	0.176
Recording Settings				
recording durations (day)	17.1 ± 20.5	20.9 ± 15.4	17.8 ± 19.7	0.036
time since implant (day)	12 ± 34	18 ± 40	13 ± 35	0.792
sensing frequency (Hz)	17.2 ± 13.0	16.4 ± 3.9	17.1 ± 11.9	0.018

M: Male, F: Female, TEED: total electrical energy delivered, LEDD: levodopa equivalent daily dosage

Table 2. Generalized Linear Model for effect of stimulation therapy and medications in Beta-band (13-30Hz) subgroup.

	Coefficient	Std. Err.	z	P>|z|	[0.025	0.975]
const	0.8452	0.537	1.575	0.115	-0.207	1.897
hemisphere	0.1552	0.245	0.633	0.526	-0.325	0.635
channel	-0.4221	0.28	-1.508	0.132	-0.971	0.126
LEDD	-0.0004	0	-1.78	0.075	-0.001	4.42E-05
levodopa ER	0.76	0.269	2.825	**0.005**	0.233	1.287
benzo	-0.1727	0.257	-0.673	0.501	-0.676	0.33
melatonin	-0.113	0.272	-0.416	0.678	-0.646	0.42
TEED	-0.0307	1.154	-0.027	0.979	-2.293	2.232

TEED: total electrical energy delivered, LEDD: levodopa equivalent daily dosage

Figures

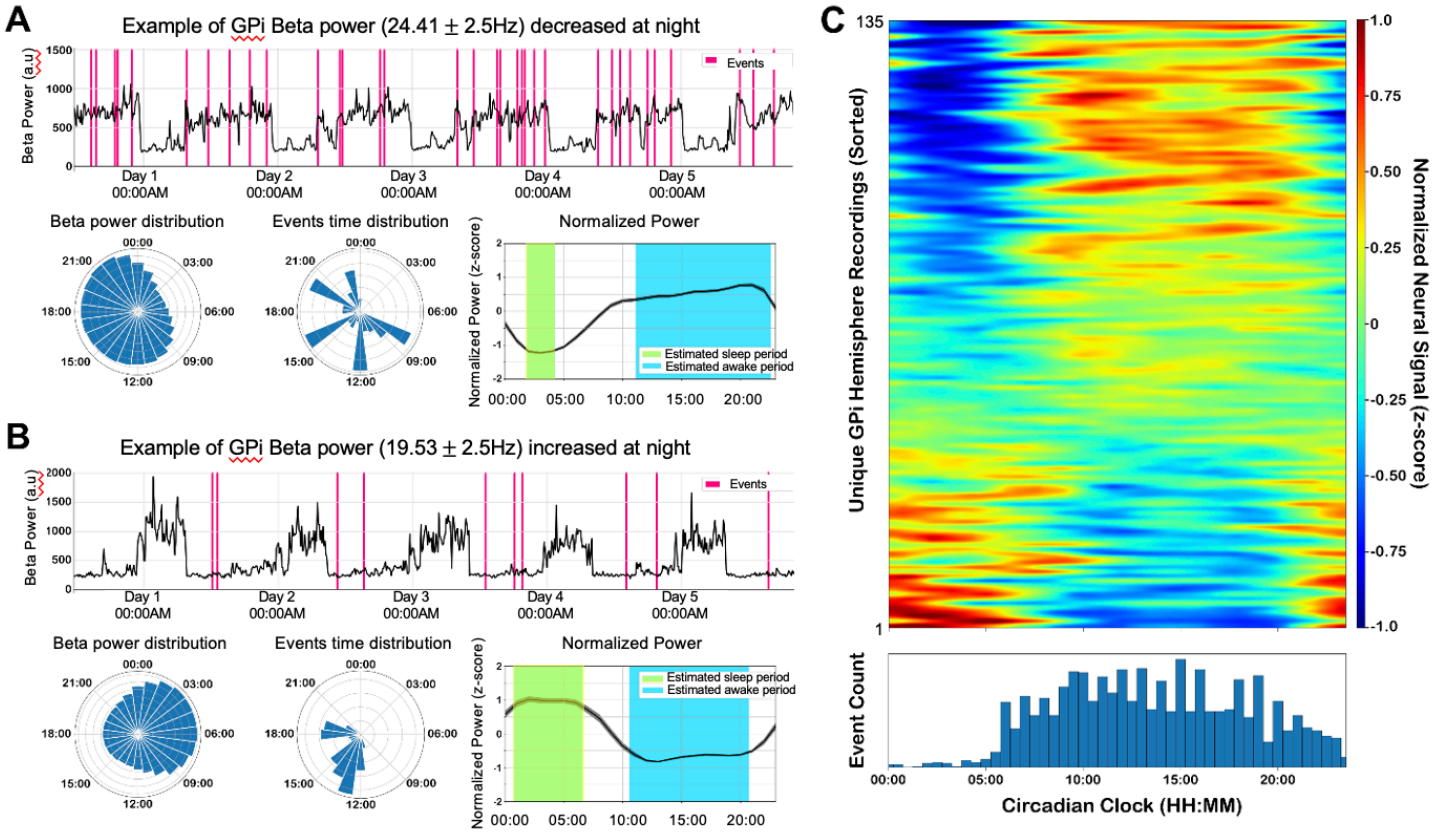

Figure 1

Pallidal circadian rhythm. A) Example of beta GPi activity (24.41 ± 2.5Hz) recorded over 5 days and decreasing at night. Pink vertical lines indicate the time of events marked by patient and used here as a surrogate marker of wakefulness. Circular polar plots show the beta power (left panel) and number of events marked (middle panel) in a 24-hour circadian clock with 1-hour averaged increment. Normalized neural signals averaged over 24-hour cycle with the estimated sleep and awake period (see method, right panel). **B**) Example GPi beta band (19.53 ± 2.5Hz) increasing at night. Same convention as in panels A. **C**) Circadian heatmap showing each individual GPi recordings (n=135) normalized and plotted over 24-hour circadian clock. GPi power is sorted from the most decreased (blue) to the most increase (rest) at

night. The number of events reported by all subjects over 24-hour circadian clock is shown below the circadian heatmap and used as a surrogate for awake/sleep period.

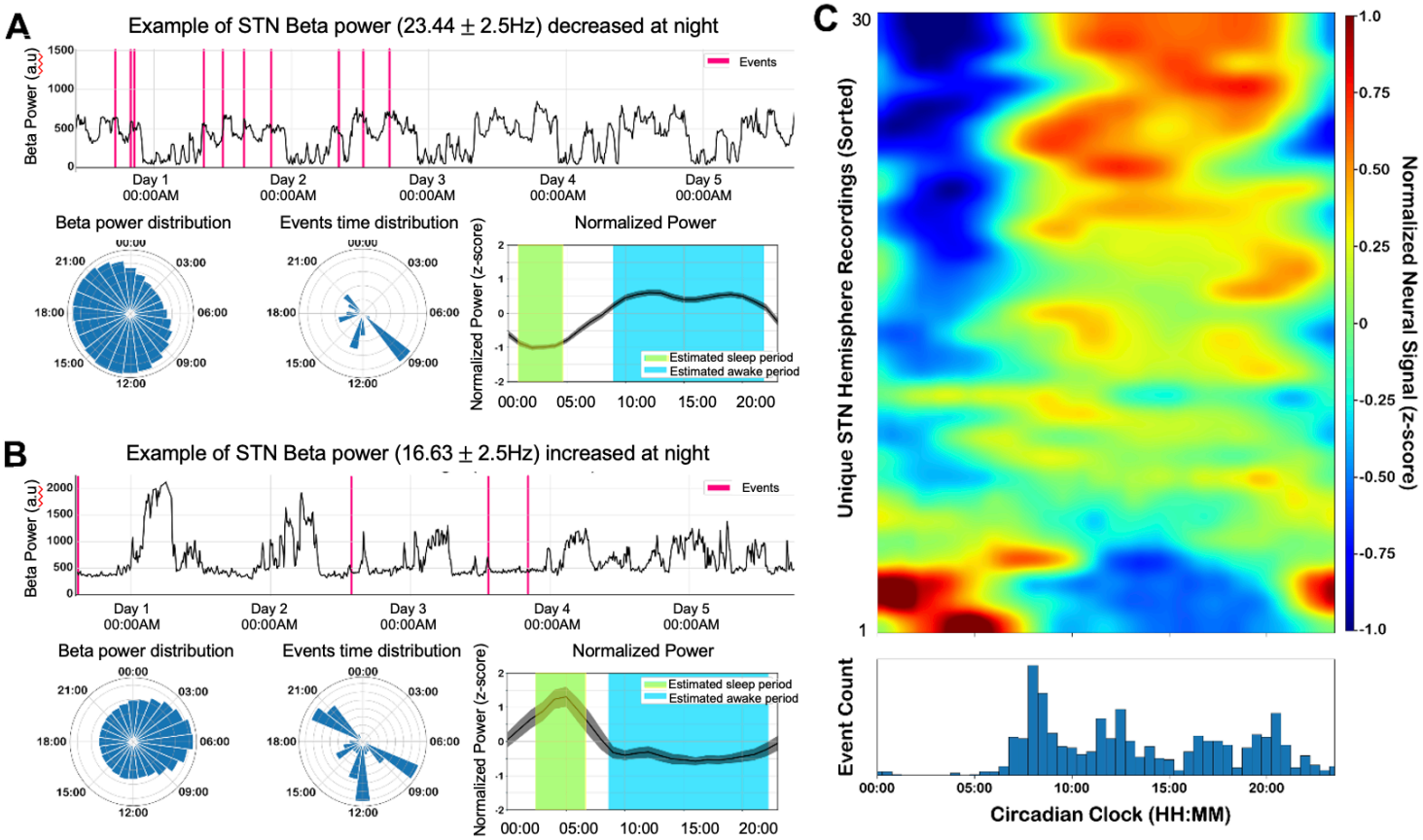

Figure 2

STN circadian rhythm. A) Example of beta STN activity (23.44 ± 2.5Hz) recorded over 5 days and decreasing at night. Pink vertical lines indicate the time of events marked by patient and used here as a surrogate marker of wakefulness. Circular polar plots show the beta power (left panel) and number of events marked (middle panel) in a 24-hour circadian clock with 1-hour averaged increment. Normalized neural signals averaged over 24-hour cycle with the estimated sleep and awake period (see method, right panel). **B**) Example STN beta band (16.63 ± 2.5Hz) increasing at night (. Same convention as in panels A. **C**) Circadian heatmap showing each individual STN recordings (n=30) normalized and plotted over 24-hour circadian clock. STN power is sorted from the most decreased (blue) to the most increase (rest) at night. The number of events reported by all subjects over 24-hour circadian clock is shown below the circadian heatmap and used as a surrogate for awake/sleep period.

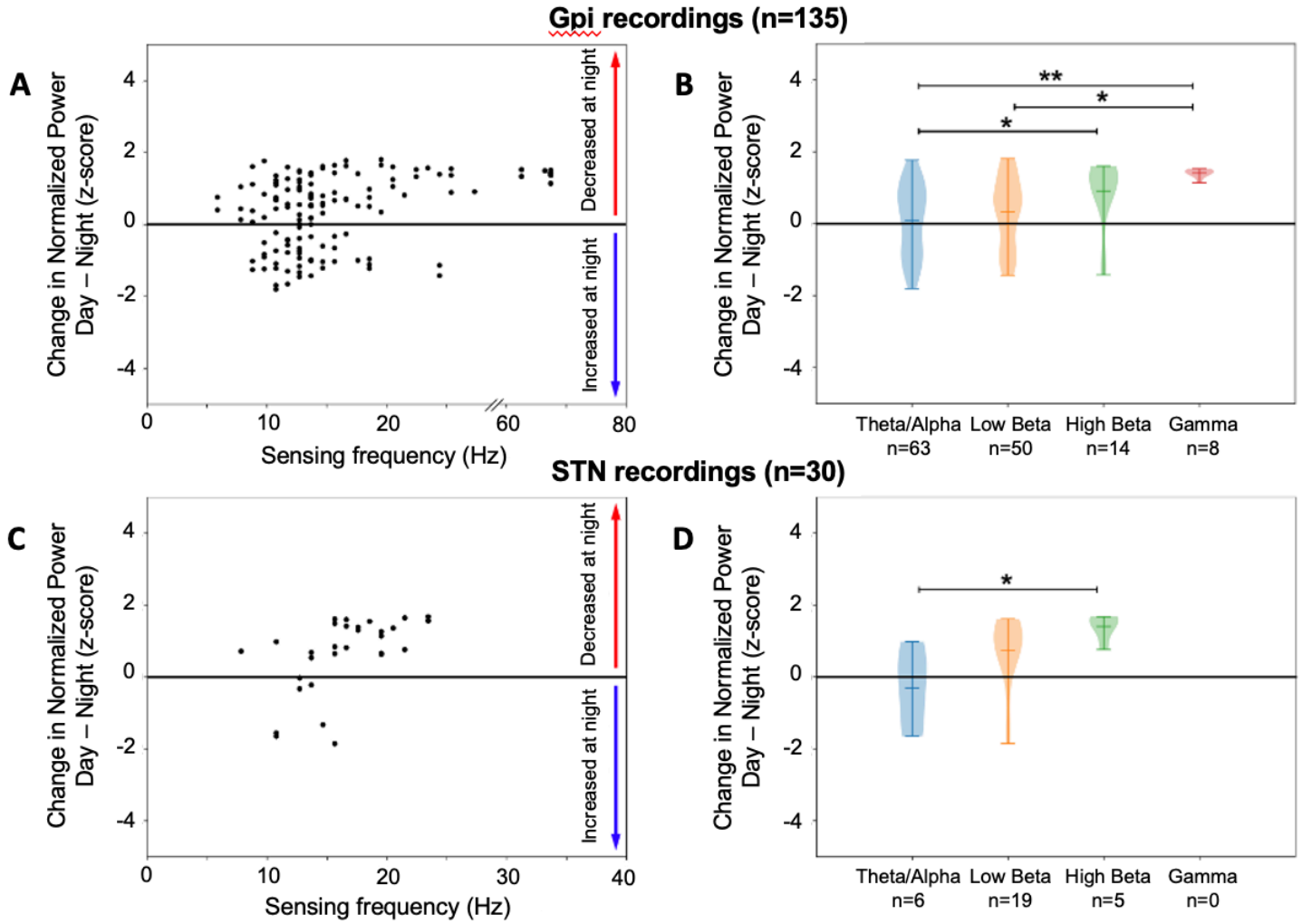

Figure 3

Figure 2. Circadian rhythm depends on the frequency of signal tracked. A) Scatter plot of the changes in neural signals between awake and sleep period (Day – Night), for each GPi hemisphere recordings (n=135), in relationship to the center frequency of the recordings. A positive change indicates a reduction of power at night while a negative value indicates an increase in power at night. **B**) Violin plot quantifying the changes in power during day and night for each canonical frequency bands; Theta/Alpha (<12Hz), Low-beta (13-20Hz), High-beta (21-30Hz), and Gamma (>31Hz). High-beta (p=0.033) and Gamma (p=0.031) distributions were statistically more likely to decrease at night (indicated by positive values and * below violins) while Theta/Alpha and Low-beta were not statistically different from 0, indicating that power was either increased or decreased at night in those frequency bands. Gamma band was statistically different from Theta/Alpha (p=0.002) and Low-beta (p=0.019) band, while High-beta band was statistically different from Theta/Alpha band (p=0.023). **C** and **D.** same convention as A. and B but for STN recordings (n=30)

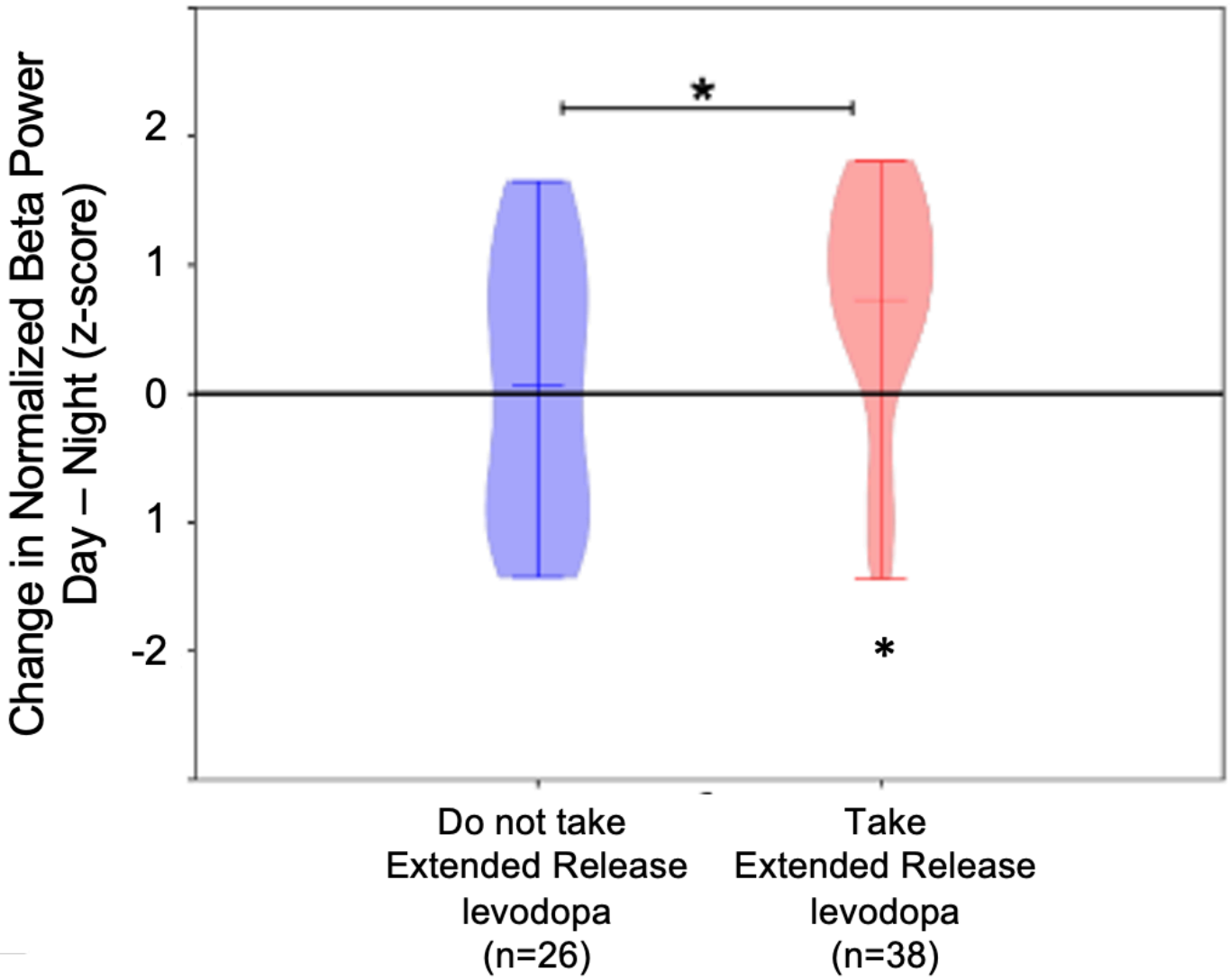

Figure 4

Figure 3. Extended-release medication modulates circadian Pallidal Beta rhythm. Comparison of circadian Beta power in subjects treated with Levodopa extended-release (red violin) and those who do not take extended-release (blue violin). In subjects taking extended-release Beta power was more likely to decrease at night, with a distribution statistically different from 0 ($p < 0.0001$) and was different from subjects who were not treated with extended-release medication ($p = 0.016$).

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- SupplementaryFigure1.docx

- NCOMMS2350065rs.pdf